

Protechsoft TypeScript Notes day 1

1. Introduction to TypeScript

TypeScript (TS) is a superset of JavaScript developed by Microsoft.

It adds ****static typing**** to JavaScript.

TypeScript code is converted (compiled) into JavaScript before running in the browser.

Advantages of TypeScript

- * Type Safety
- * Better Code Quality
- * Easier Debugging
- * Better IntelliSense Support
- * Supports OOP Features
- * Detects Errors During Compilation

Questions

****Q1. What is TypeScript?****

Ans: TypeScript is a superset of JavaScript that adds static typing.

****Q2. Who developed TypeScript?****

Ans: Microsoft.

****Q3. Can browsers run TypeScript directly?****

Ans: No. TypeScript must be compiled into JavaScript.

2. Installation

Install TypeScript Globally

```
npm install -g typescript
```

Check Version

```
tsc -v
```

3. How to Compile and Run TypeScript

Create File

```
app.ts
```

Compile

```
tsc app.ts
```

This generates:

```
app.js
```

Run

```
node app.js
```

Auto Compile

```
tsc --watch
```

or

```
tsc -w
```

Questions

****Q1. Which command compiles TypeScript?****

Ans:

bash

tsc filename.ts

****Q2. Which command continuously watches changes?****

Ans:

bash

tsc --watch

4. Data Types

Data types specify what kind of value a variable can store.

5. Type Annotation

Type annotation is the process of explicitly specifying a datatype.

Syntax

variableName : datatype

Example

```
let age : number = 25;
```

Question

****Q. What is Type Annotation?****

Ans: Explicitly specifying the datatype of a variable.

6. Number Data Type

Used to store numeric values.

Example

```
let a : number;
```

```
a = 10;
```

```
console.log(a);
```

Invalid Example

```
a = "hello"; // Error
```

7. String Data Type

Used to store text values.

Example

```
let username : string;
```

```
username = "Virat";
```

Invalid Example

```
username = true; // Error
```

8. Boolean Data Type

Stores true or false values.

Example

```
let isPlayer : boolean;
```

```
isPlayer = true;
```

```
isPlayer = false;
```

Invalid Example

```
isPlayer = "hello"; // Error
```

9. Special Data Types

any

The ``any`` datatype can store any type of value.

Example

```
let a : any;
```

```
a = "hello";
```

```
a = 10;
```

```
a = true;
```

Drawback

TypeScript performs no type checking.

```
let a : any;  
  
a = 10;  
  
console.log(a.toUpperCase());
```

Runtime Error ✕

Question

****Q. What is any?****

Ans: A datatype that disables type checking.

unknown

The ``unknown`` datatype can store any value, but type checking is required before use.

Example

```
let b : unknown;  
  
b = "hello";  
  
if(typeof b === "string"){  
    console.log(b.toUpperCase());  
}
```

Benefits

- * Safer than any
- * Prevents accidental errors

Question

****Q. Difference between any and unknown?****

any	unknown	
-----	-----	
No type checking	Requires type checking	
Unsafe	Safe	

void

Used when a function does not return any value.

Example

```
function add() : void{  
    console.log("Hello");  
}
```

Question

****Q. What is void?****

Ans: Used when a function does not return anything.

never

Used when a function never completes execution.

Example

```
function counting() : never{  
    while(true){  
    }  
}
```

Uses

- * Infinite loops
- * Functions that throw errors

Question

****Q. What is never datatype?****

Ans: Used when a function never returns a value.

10. Arrays

Array is a collection of similar datatype elements.

String Array

```
let names : string[] = [  
    "Virat",  
    "Rohit",  
    "Iyer"  
];
```

Add Element

```
names.push("Dhoni");
```

Number Array

```
let marks : number[] = [  
    10,  
    20,  
    30  
];
```


Boolean Array

```
let booleanArr : boolean[] = [  
    true,  
    false  
];
```

11. Nested Arrays

Array inside another array.

String Nested Array

```
let nested : string[][] = [  
    ["HTML", "CSS", "JS"],  
    ["Java", "Python", "SQL"]  
];
```

Add New Array

```
nested.push([  
    "Node",  
    "Express",  
    "MongoDB"  
]);
```

Number Nested Array

```
let nested2 : number[][] = [  
    [10, 20, 30],  
    [40, 50, 60]  
];
```

12. Traversing Arrays

Using For Loop

```
for(let i=0;i<names.length;i++){  
    console.log(names[i]);  
}
```

Output

```
text  
Virat  
Rohit  
Iyer
```

Using map()

Used to traverse and create a new array.

Example

```
let players = names.map((ele)=>{  
    return ele.toUpperCase();  
});
```

Output

```
["VIRAT","ROHIT","IYER"]
```

Features

- * Returns a new array
- * Does not modify original array